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New Theoretical Perspectives on Molecular Shape Resonances: Feshbach-Fano Methods for Mulliken Orbital Analysis of Photoionization Continua*

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Abstract

Recent progress is reported in theoretical, computational, and semiempirical studies of shape resonances in the partial photoionization cross sections of diatomic and polyatomic molecules. A multi-channel Feshbach-Fano formalism is described which is particularly well suited for the analysis of resonance ionization channels. The development allows calculations in non-degenerate point-group-symmetry scattering waves, provides compact expressions for partial-channel cross sections, and gives generalized Fano-like absorption lineshapes of parametrizable form. An appropriately selected Q space of zeroth-order molecular configurational states, corresponding to the (σ, σ^*) charge-transfer excitations of Mulliken, is seen to account for the appearance of prominent features in the cross sections of simple molecules. These zeroth-order states further provide a basis for understanding and elaborating upon semi-empirically determined correlations between bond lengths and resonance energies in the valence-and K-shell cross sections of linear and quasi-linear molecules. A universal correlation is derived theoretically in these cases which relates resonance energies to effective bond lengths, square-well quantum numbers, and effective potentials, each of which is obtained *a priori* from elementary considerations. Recent studies of C_2H_2, C_2H_4 and C_2H_6 photoionization cross sections illustrate the complexities that can arise in compounds which exhibit two σ^* resonances (C–C and C–H) in some channels. Mulliken configurational states underlying resonance features in these compounds are seen to account satisfactorily for the natures of calculated and measured partial cross sections in the absence of simple square-well correlation behaviour. Connections of the development with conventional computational methods based on K-matrix, S-matrix, or eigenchannel states are indicated, and cautionary remarks are made on the use of well models in describing photoionization resonances.

1. Introduction

Experimental studies of valence- [1] and K-shell [2] photoionization processes in gas-phase [3] and adsorbate [4] compounds provide complementary information on initial- and final-state molecular electronic wavefunctions under differing circumstances. Recent empirical and semi-empirical investigations attempt to correlate observed resonance positions [5] and lineshapes [6, 7] with molecular geometry and with the influence of adsorption. Corresponding theoretical developments based on Feshbach–Fano methods [8–12] can provide *ab initio* absorption lineshapes of parametrizable form, help to specify unambiguously the natures and origins of prominent cross sectional features [13–15], and identify the possible origins of the correlations inferred semi-empirically [16, 17].

In the present report, recently devised theoretical methods are described which are particularly well suited for point-group-symmetry descriptions of molecular photoionization cross sections [8–12], and applications are reported to the study of the origins and natures of prominent resonance features which appear in certain photoionization channels of diatomic and polyatomic molecules [13–17]. Absorption lineshapes of parametrizable form and approximate correlations of resonance spectral poritions with bond lengths are seen to follow directly from the theoretical development upon identification of Mulliken (σ, σ^*) charge-transfer configurations as zeroth-order states underlying the prominent spectral features.

In Section 2, so-called prepared states of point-group-symmetry type [8–12] are introduced to resolve the infinite degeneracy associated with the photoionization continuum. Procedures for the calculation of cross sections are reported in Section 3 employing non-degenerate prepared states and the P-Q projection-operator formalism of Feshbach [18]. A Q-space of Mulliken (σ, σ^*) charge-transfer configurational states constructed in minimal-basis-set virtual valence orbitals in seen to account for cross-sectional features which are generally designated as shape resonances on basis of scattering wave calculations in familiar cases (N_2, CO_2) [19, 20]. Lineshapes of generalized Fano type [21, 22], which incorporate Lorentzian limits and can be parametrized if desired, are obtained for channels containing one or more resonance features.

In Section 4, a universal correlation between resonance positions and molecular bond lengths in valence-and K-shell ionization cross sections is derived for linear and quasi-linear compounds on basis of the natures and behaviors of Mulliken states underlying the shape-resonance features [15–17]. The theoretically derived correlation is seen to account for experimentally and computationally determined resonance positions, thereby clarifying the origin of related semi-

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empirically derived correlations and helping to establish their range of validity [5].

Calculations on simple hydrocarbons (C_2H_2, C_2H_4, C_2H_6) presented in Section 5 illustrate cases in which two different (C–C and C–H) types of σ^* antibonding orbitals are present in the same compound. Zeroth-order Mulliken configurational states are seen to account for the origins and natures of prominent features in the observed cross sections in these cases in the absence of simple square-well correlation behavior. The variation of (σ, σ^*) configurational state energy with both C–C bond length and with the number of C–H bonds present is seen to be of importance in accounting for the positions of the observed and calculated cross-sectional maxima in these compounds.

Concluding remarks and comparisons with alternative perspectives on calculations and interpretations of molecular shape resonances, on their correlations with molecular geometry, and on ionization lineshapes are made in Section 6.

2. Prepared-state development

Partial-channel photoionization cross sections are generally written in the form of a sum over unresolved degenerate-state contributions which can be associated with the directions of ejected photoelectrons or with other unresolved state specifications [23].

$$\sigma_{i\to t}(E) = \sum_{\alpha} |\langle \Phi_E^{(\alpha)} | \boldsymbol{\mu}\cdot\boldsymbol{\varepsilon} | \Phi_i \rangle|^2. \quad (1)$$

Here, $\Phi_E^{(\alpha)}$ is a vibronic continuum Schrödinger solution at energy E, t specifies a particular energetic channel labelled by appropriate final ionic-state vibronic quantum numbers, α provides the degenerate state labelling, generally associated with the direction of photoejected electron, i refers to the initial state of the molecule, $E - E_i = h\nu$ is the quantum energy of incident light of frequency ν, having polarization direction given by the unit polarization vector $\boldsymbol{\varepsilon}$, and $\boldsymbol{\mu}$ is the dipole moment operator. The continuum states $\Phi_E^{(\alpha)}$ satisfy [24]

$$\Phi_E^{(\alpha)} = \phi_E^{(\alpha)} + G(E)V\phi_E^{(\alpha)}, \quad (2)$$

where $\phi_E^{(\alpha)}$ is a known reference solution for the energetic channel t of interest and $G(E)$ is the full Green's function chosen to satisfy the appropriate asymptotic conditions. Familiar degenerate state labels (α) include; (i) the asymptotic linear momenta ($\boldsymbol{k}$) of directed-wave solutions satisfying incoming radial wave boundary conditions associated with S-matrix normalized states [25], (ii) the asymptotic angular momenta of l-wave solutions associated with K-matrix normalized states [26], and (iii) the eigenchannel values associated with the degenerate components of the irreducible symmetries of the scattering potential [27]. All such degenerate continuum states are conveniently Dirac delta-function normalized on the energy scale [28].

Rotational degrees of freedom are treated classically in eq. (1) [29], which equation consequently requires an angle average over orientations, in accordance with implied summations over nearly degenerate rotational states in a quantum description of the differential cross section [30]. Such vibronic cross sectional expressions are also appropriate for oriented adsorbed molecules, in which cases the angle average refers only to possible bending motions of the adsorbate relative to the bonding surface. The electronic portions of the states of eqs. (1) and (2) can be taken in both gas-phase and adsorbate cases as (Hund's case b) point-group-symmetry states for the circumstances of interest [12].

The (Fano-Cooper) dipole-prepared states [21]

$$\Phi_E^{(\mathrm{d})} = \sum_{\alpha} \Phi_E^{(\alpha)} \langle \Phi_E^{(\alpha)} | \boldsymbol{\mu}\cdot\boldsymbol{\varepsilon} | \Phi_i \rangle \quad (3)$$

resolve the infinite continuum degeneracy in the channel $i \to t$, and provide a convenient cross-sectional expression of the form

$$\sigma_{i\to t}(E) = \langle \Phi_i | \boldsymbol{\mu}\cdot\boldsymbol{\varepsilon} | \Phi_E^{(\mathrm{d})} \rangle. \quad (4)$$

Equations (3) and (4) are most useful when the dipole-prepared states can be calculated without prior reference to the individual component $\Phi_E^{(\alpha)}$ of which they are comprised. These dipole-prepared states are calculated directly in the body frame in the present development for the three appropriate principal-axis polarization directions following the procedures outlined below, and described in detail elsewhere [8–17]. An angle average of eq. (4) is required over the known or assumed molecular orientational distribution in the case of both gas-phase and adsorbate targets in order to obtain cross sections appropriate for comparison with experiment, as indicated above.

3. Feshbach–Fano development

The partial-channel cross sections of eqs. (1) and (4) can include prominent features associated with Rydberg, Feshbach, or shape-resonance states, giving rise to continuum functions which exhibit significantly different spatial characteristics at different but closly adjacent energies. It is convenient in such cases to adopt Feshbach–Fano methods which separate continuum spectra into resonant and background contributions at the outset and provide lineshape functions which include physically significant width, shift, and other energy-dependent parameters [18, 21, 22]. Specifically, the projector

$$Q = \sum_j |j\rangle\langle j| \quad (5)$$

is employed, where the L^2 vectors $|j\rangle$ are largely arbitrary, but are chosen here to represent in the zeroth order possible shape-resonant states present in the continuum channel ($i \to t$) of interest. The particular choice of such zeroth-order states $|j\rangle$ adopted for molecular photoionization studies is described in further detail below.

Introducing the complement projector $P = 1 - Q$, the channel cross section is obtained in the form [10, 11]

$$\sigma_{i\to t}(E) = \sigma_{\mathrm{b}}(E) - (1/\pi)\,\mathrm{Im}\,[\boldsymbol{g}(E)\cdot\boldsymbol{G}_{\mathrm{r}}(E)\cdot\boldsymbol{g}(E)] \quad (6)$$

where

$$\sigma_{\mathrm{b}}(E) = \langle \chi_E^{(\mathrm{d})} | \boldsymbol{\mu}\cdot\boldsymbol{\varepsilon} | \Phi_i \rangle, \quad (7)$$

is a nonresonant background cross section, and

$$\boldsymbol{g}(E) = \boldsymbol{\rho}(E) - \mathrm{i}\pi\boldsymbol{q}(E), \quad (8)$$

is the profile vector, with $\boldsymbol{\rho}(E)$ and $\boldsymbol{q}(E)$ providing vector generalizations of the ρ and q parameters of Fano [21, 22] in the forms

$$\rho_j(E) = \langle \Phi_i|\boldsymbol{\mu}\cdot\boldsymbol{\varepsilon}|\chi_j^{(\mathrm{H})}\rangle = \langle \chi_E^{(\mathrm{d})}|H'|j\rangle \quad (9)$$

$$q_j(\mathrm{E}) = \langle \Phi_i|\boldsymbol{\mu}\cdot\boldsymbol{\varepsilon}|\chi_j^{(\mathrm{M})}\rangle. \quad (10)$$

In eqs. (7) to (10),

$$\chi_E^{(\mathrm{d})} = \sum_\alpha \chi_E^{(\alpha)}\langle\chi_E^{(\alpha)}|\boldsymbol{\mu}\cdot\boldsymbol{\varepsilon}|\Phi_i\rangle \quad (11)$$

$$\chi_j^{(\mathrm{H})} = \sum_\alpha \chi_E^{(\alpha)}\langle\chi_E^{(\alpha)}|H'|j\rangle + \sum_n |n\rangle\langle n|H'|j\rangle\delta(E-E_n),$$

$$j = 1, 2, \ldots, n, \quad (12)$$

and

$$\chi_j^{(\mathrm{M})} = |j\rangle + \int \mathrm{d}E'\chi_j^{(\mathrm{H})}/(E-E') \quad (13)$$

are dipole-prepared, energy-prepared, and energy-modified states, respectively. These are expressed as indicated in terms of the background channel $(i\to t)$ functions $\chi_E^{(\alpha)}$ satisfying $(PHP - E)\chi_E^{(\alpha)} = 0$, where $H' = PHQ + QHP$ is the interaction Hamiltonian, along with E_n, $|n\rangle$, the bound eigenvalues and functions of PHP, and the L^2 states $|j\rangle$ of the projector Q in eq. (5).

The matrix $\boldsymbol{G}_\mathrm{r}(E)$ in eq. (6) represents the resonant portion $[QG(E)Q]$ of the full propagator, and can be written [10, 11]

$$\boldsymbol{G}_\mathrm{r}(E) = [E\boldsymbol{I} - \boldsymbol{H}_Q - \Delta(E) + \mathrm{i}\Gamma(E)/2]^{-1}, \quad (14)$$

with $\boldsymbol{H}_Q$ the Q-space Hamiltonian matrix, and

$$\Gamma_{ij}(\mathrm{E}) = 2\pi\langle i|H'|\chi_j^{(\mathrm{H})}\rangle = 2\pi\langle\chi_i^{(\mathrm{H})}|H'|j\rangle \quad (15)$$

$$\Delta_{ij}(E) = (2\pi)^{-1}\int \mathrm{d}E'[\Gamma_{ij}(E')]/(E - E'), \quad (16)$$

providing width and shift matrix elements, respectively. Cauchy principal values of the singular integrals of eqs. (13) and (16) are understood.

The foregoing development provides a complete prescription for determination of the channel cross section $\sigma_{i\to t}(E)$ of eq. (6) in terms of $\sigma_\mathrm{b}(E)$, $\rho_i(E)$, $q_i(E)$, $\Gamma_{ij}(E)$, and $\Delta_{ij}(E)$, which are obtained from the indicated prepared states of eqs. (3), (11), and (12) following L^2 computational procedures described in detail elsewhere [8–13]. The previously reported development employs L^2 scattering solutions of the form [8–13]

$$\Phi_E^{(n)} = \sum_{j=1}^{n} q_j(E)q_j(A)|\Phi\rangle, \quad (17)$$

where the $q_j(E)$ are polynominals determined recursively, $\boldsymbol{A}$ is a matrix representation of the Hamiltonian or other appropriate operator, and $|\Phi\rangle$ is any one of the test functions defining the prepared states of eqs. (3), (11), and (12). It is perhaps appropriate to note that use of such point-group-symmetry prepared states in the present development avoids the l-wave representations employed in other treatments of complex molecular spectra, consequently providing somewhat more compact expressions than are generally obtained [31].

In the special case that Q contains a single state, the cross section of eqs. (6) to (16) takes the familiar Fano form [21, 22]

$$\sigma_{i\to t}(E) = \sigma_\mathrm{b}(E) + \frac{[q(E)/\pi + \varepsilon(E)\rho(E)]^2}{[1 + \varepsilon(E)^2]\cdot[\Gamma(E)/2\pi]} - \frac{\rho(E)^2}{[\Gamma(E)/2\pi]} \quad (18)$$

where

$$\varepsilon(E) = [E - E_Q - \Delta(E)]/[\Gamma(E)/2] \quad (19)$$

is a scaled energy variable, and $\rho(E)$, $q(E)$, E_Q, $\Delta(E)$, and $\Gamma(E)$ are defined in eqs. (6) to (16). In the present development employing interaction-prepared states only a single nondegenerate continuum molecular state is required at each energy E in determinations of these parameters. By contrast, when an l-wave representation is employed in studies of molecular continua, a multi-channel formalism in required even for a single energetic open channel $(i\to t)$. Moreover, eq. (6) is parametrizable under appropriate conditions in terms of the vector of eqs. (8) to (10) and the resonance matrix of eqs. (14) to (16). The latter requires off-diagonal shift and width elements which must be determined iteratively in an appropriate fitting procedure which is beyond the scope of interest of the present report.

Finally, when the single resonance state is chosen to be the so-called dipole test function $Q \to \Phi = (\boldsymbol{\mu}\cdot\boldsymbol{\varepsilon})\Phi_i$, the Fano profile of eq. (18) takes the Lorentzian form

$$\sigma_{i\to t}(E) = \frac{2|\langle\Phi|\Phi\rangle|^2/[\pi\Gamma(E)]}{1 + \varepsilon(E)^2} \quad (20)$$

where the background cross section is identically zero and calculation of $q(E)$ and $\rho(E)$ have been obviated [8, 9]. The width and shift functions are obtained in this case from Stieltjes calculations which are in every way identical to those employed for nonresonant channels [12]. This and additional aspects of employing eq. (20) in L^2 Freshbach–Fano treatments of molecular shape resonances have been discussed previously [8, 9].

In order to identify appropriate projector states, illustrative static-exchange calculations are reported here in the cases of the familiar $3\sigma_\mathrm{g}$ and $4\sigma_\mathrm{g}$ photoionization channels in N_2 and CO_2, respectively. The development employs matrix representations of the Hamiltonian operator of eq. (17) constructed in Gaussian basis sets of approximaly 225 and 245 terms, respectively, and single Q-space projectors comprised of Mulliken charge-transfer (σ, σ^*) states [$(3\sigma_\mathrm{g}, 3\sigma_\mathrm{u})$ and $(4\sigma_\mathrm{g}, 4\sigma_\mathrm{u})$, respectively] in each case [13]. Such states have long been suspected of giving rise to molecular photoionization resonances [32], although projection calculations of the type reported here have been lacking. Details relating to the specific basis sets adopted, to other computational aspects of the results reported, and to comparisons between theory and experiment are given elsewhere [11].

Figure 1 compares calculated partial cross sections for the (a) $3\sigma_\mathrm{g} \to k\sigma_\mathrm{u}$ and (b) $4\sigma_\mathrm{g} \to k\sigma_\mathrm{u}$ channels in N_2 and CO_2 molecules, respectively. Evidently, the background cross sections obtained in both cases (Fig. 1) are small and smooth when these (σ, σ^*) states are employed in the Q space, suggesting that the latter can indeed be regarded as responsible for the resonances present in the channel cross sections. The N_2 background cross section, in particular, is seen to be essentially constant over the width of the resonance. The parameters $\Gamma(E)$, $\Delta(E)$, $\rho(E)$, and $q(E)$ for these two channels, shown in Fig. 2, are also generally smooth functions of the energy. Since the convention of Fano [21] has been adopted in defining the $\rho(E)[\to\rho/(\sigma_\mathrm{b}\Gamma/2\pi)]$ and $q(E)[\to q/\rho\pi]$ parameters shown in Fig. 2, $q(E)$ is singular for the $4\sigma_\mathrm{g}$ CO_2 channel at approximately 28 eV, where the corresponding $\rho(E)$ goes through zero. Of course, the cross section is a continuous function of energy over the entire spectral range shown. A zero value for $\rho(E)$ in this case implies that the

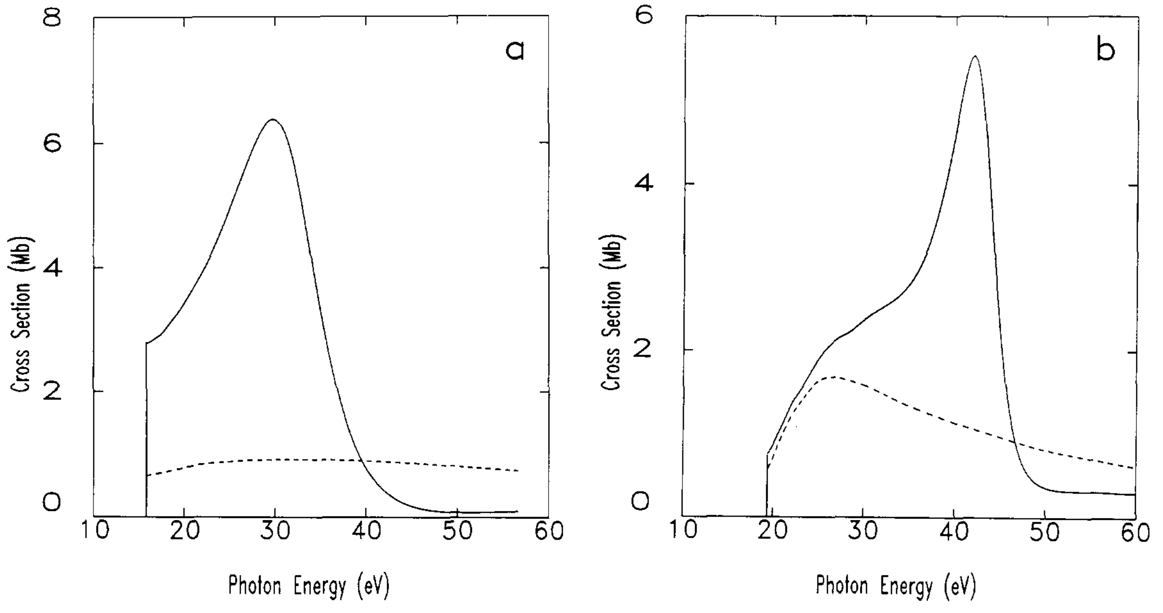

Fig. 1. Background (- - -) and channel (——) cross sections for (a) $3\sigma_g \to k\sigma_u$ and (b) $4\sigma_g \to k\sigma_u$ ionization in N_2 and CO_2 molecules, respectively, calculated employing Feshbach–Fano methods as described in Section 3 of the text, and (σ, σ^*) configurational-state representations of the resonances in zeroth-order [9–11]. The reported channel cross sections are in good accord with the result of scattering calculation employing partial-wave expansions and numerical integration techniques [19, 20]. Corresponding lineshape parameters are shown in Fig. 2.

dipole- and energy-prepared states of eqs. (11) and (12), respectively, are orthogonal and refer to two entirely different linear combinations of degenerate continuum states at this energy. These illustrative results indicate that in some cases [e.g., N_2] a parametrized generalized Fano lineshape can be employed to represent shape resonance profiles, since the appropriate parameters are largely energy independent. By contrast, in other cases [e.g., CO_2] more significant energy dependence will be evident, arguing against an energy-independent parametrization.

In Fig. 3 are shown the results of single-channel static-exchange calculations for the (a) $1\sigma_g \to k\sigma_u$ and (b) $3\sigma_g \to k\sigma_u$ ionization channels in molecular nitrogen, obtained employing the Feshbach-Fano development of Section 3 and an identical $3\sigma_u(\sigma^*)$ virtual orbital in the appropriate zeroth-order configurational states in each case. Of course, the various lineshape parameters (not shown) of eqs. (7) to (16) differ in the two cases, since the two sets of background channel function of eqs. (11) and (12) are different. It is seen from the figure that smooth and unstructured background cross sections are obtained, indicating the same σ^* virtual orbital is responsible for the shape resonances in both the valence- and K-shell channels in N_2. Note that an unrelaxed molecular-orbital or delocalized-hole state ($1\sigma_g^{-1}$) is employed in the K-edge calculation. Use of relaxed or local-hole states can result in quantitatively different K-edge cross sections, in

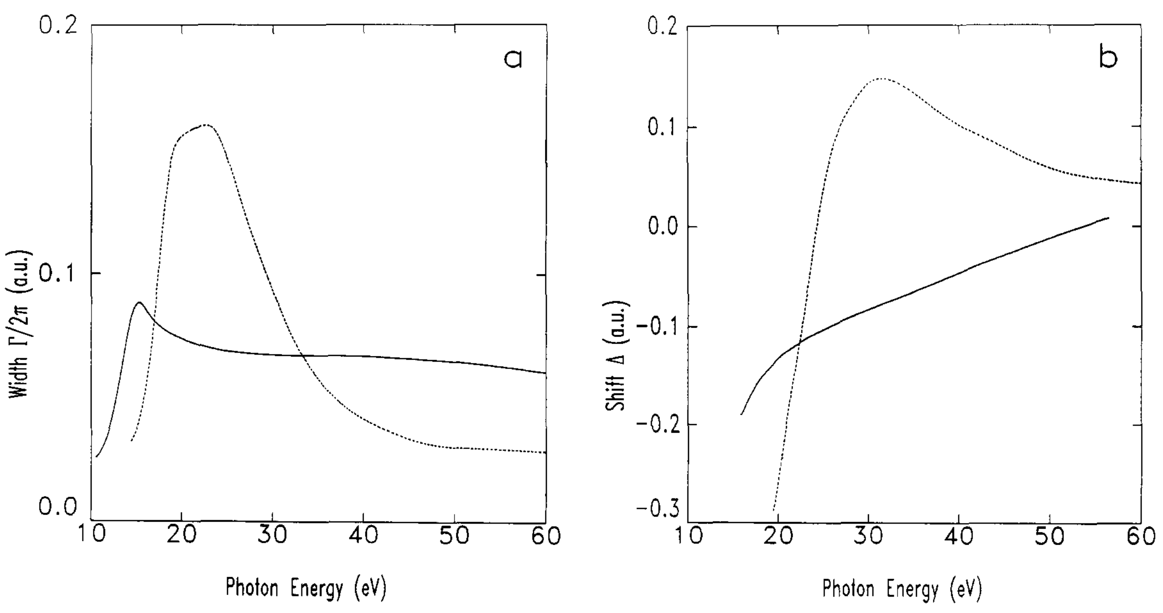

Fig. 2. Lineshape parameters (a) $\Gamma(E)$, (b) $\Delta(E)$, (c) $\rho(E)$, and (d) $q(E)$, of Eqs. (7) to (16) for the N_2 (——) and CO_2 (····) channel cross sections of Figure 1, obtained following procedures described in Section 3 of the text. Matrix representations of dimensions ~ 225 and 245, respectively, are employed for the appropriate operators, and moment orders of ~ 10–15 are found to give convergent values for the lineshape parameters reported in the figures [9–11].

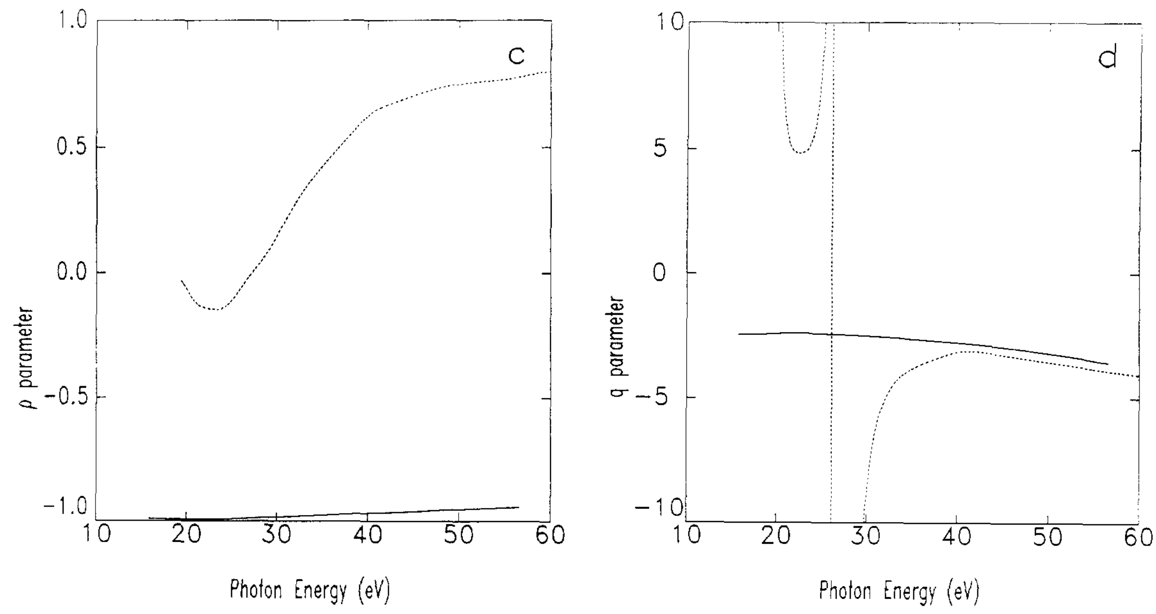

Fig. 2. Continued

which cases Mulliken charge-transfer configurations with relaxed or broken symmetry σ^* orbitals would provide the appropriate zeroth-order states. The calculations of Figs. 1 to 3 suggest the Mulliken configurational states may account for the appearance and natue of shape resonances in molecules quite generally.

4. Shape-resonance correlations with molecular geometry

The identification of Mulliken charge-transfer states underlying the resonance features in the molecular valence and K-shell photoionization cross sections of Figs 1 to 3 provides a general means for understanding the correlation of their spectral positions with molecular bond lengths, and more generally with molecular geometry [14, 15]. Additionally, the well-known similarity of valence molecular orbitals with the eigenfunctions of a particle in a one-dimensional box allows development of a simple universal relationship between energy, effective bond length, effective potential, and nodal quantum number [16]. These ideas are illustrated graphically in Fig. 4 for the occupied and virtual σ orbitals in N_2, shown enclosed in a square well of appropriate length l, chosen on basis of well-known free-electron molecular-orbital considerations [33, 34], and a depth V_0 approximately equal to the binding energy of the lowest-lying valence molecular orbital [16]. In addition to the three occupied σ valence orbitals obtained in a minimal-bases calculation in N_2, there is also one virtual $[3\sigma_u(\sigma^*)]$ orbital, which is seen to fall above the ionization threshold in this case on basis of a configurational-state or static-exchange calculation in a $3\sigma_g^{-1}$ hole state [16]. The similarities between the Fock orbitals and the eigenfunctions of a one-dimensional well are apparent from Fig. 4, and have been noted previously in various connections. It should be recognized that Fig. 4 is schematic, and does not depict

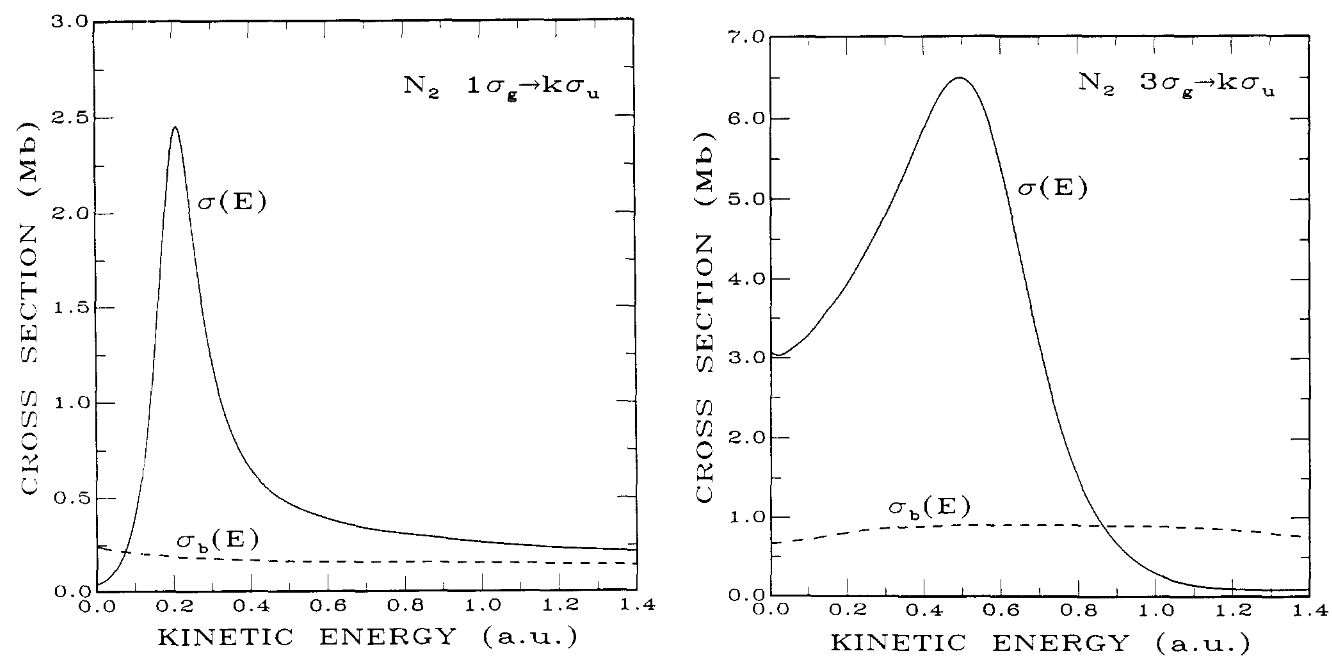

Fig. 3. Background (– – –) and channel (——) cross sections for (a) $1\sigma_g$ and (b) $3\sigma_g$ ionization in N_2, calculated employing the Feshbach–Fano development of Section 3 and identical $3\sigma_u(\sigma^*)$ virtual orbitals in each case in the zeroth-order Mulliken configurational states [16].

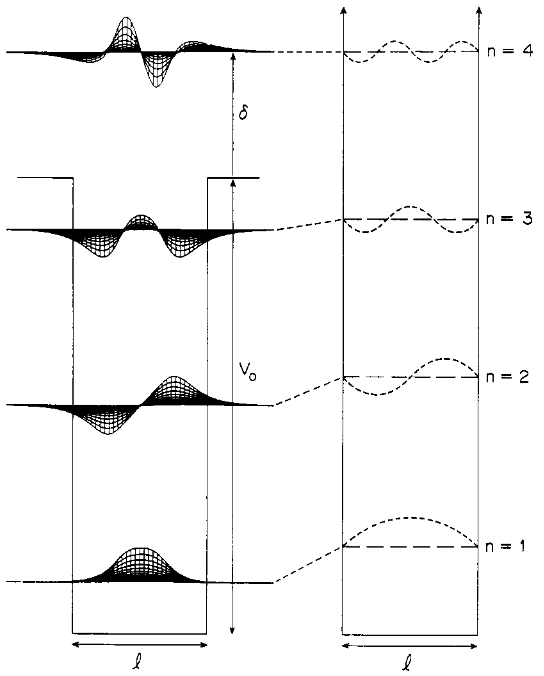

Fig. 4. One-dimensional square-well representation of the effective potential in molecular nitrogen along the symmetry axis. On the left are shown the occupied ($2\sigma_g$, $2\sigma_u$, and $3\sigma_g$) and virtual ($3\sigma_u$) σ-symmetry valence orbitals of N_2, obtained from minimal-basis-set calculations, plotted along lines parallel to but shifted off the internuclear axis by values ranging from 0.8 to 4.0 a_0. The orbitals are enclosed in a finite well of length l and depth V_0, with these parameters determined in a manner described in the text. On the right is shown an infinite well of identical length l, which provides energy levels that correlate with those of the finite well, giving a simple formula for the orbital energies measured from threshold (δ) in terms of n, l, and V_0 [see eq. (21)]. Note that the energy-level separations shown are schematic and not to scale, although the orbitals plotted refer to quantitative values.

quantative values of orbital energies, although the wave functions plotted there are taken directly from Hartree-Fock calculations, and so are quantative as shown [16].

From the well-known formula for the energy levels of an infinite square well one obtains the approximate expression [35]

$$\delta_n + V_0 = (n^2/l^2)(\pi^2/2), \tag{21}$$

where δ_n is the energy of an orbital measured from the edge of the finite well of depth V_0 as indicated in Fig. 4, l is the appropriate well length, and n is the appropriate square-well (nodal) quantum number, inferred from the spatial characteristics of the orbital of interest. Note that the results of Fig. 4 and eq. (21) apply to the symmetry axis of a cylindrical well, and not generally to a radial well, which model is less appropriate for describing the nodal behavior of an orbital along a molecular bond.

The validity of eq. (21) in providing a correlation of energy levels with molecular lengths, effective potentials, and effective quantum numbers is tested in Fig. 5(a), which reports all occupied and virtual valence-orbital energies calculated in minimal basis sets for a class of linear and quasi-linear compounds [16]. Valence-shell hole-state calculations are performed for the virtual-orbital energies in Fig. 5(a) and determinations of the appropriate square-well length and depth parameters follow the procedures for molecular nitrogen indicated above. The calculated energy levels are seen to follow closely the predictions of eq. (21), which is depicted in the figure as a dashed line havig slope ($\pi^2/2$). Figure 5(b) reports a correlation between the calculated minimal-basis-set virtual valence-orbital energies, corresponding experimental valence-shell resonance spectral positions, and the positions of cross-sectional maxima taken from corresponding single-channel valence-shell calculations [1, 16]. Evidently, the minimal-basis-set energy preditions are in good agreement with the spectral positions of experimental and theoretical cross-sectional maxima, suggesting the latter should also satisfy the simple correlation relation of eq. (21).

In Fig. 6(a) are shown the correlation of minimal-set, experimental, and theoretical valence-shell photoionization resonance energies (δ) with $1/l^2$ for linear and near linear compounds. Also shown in the figure as a dashed line of slope $\pi^2/2$ is the prediction of the square-well correlation relation-

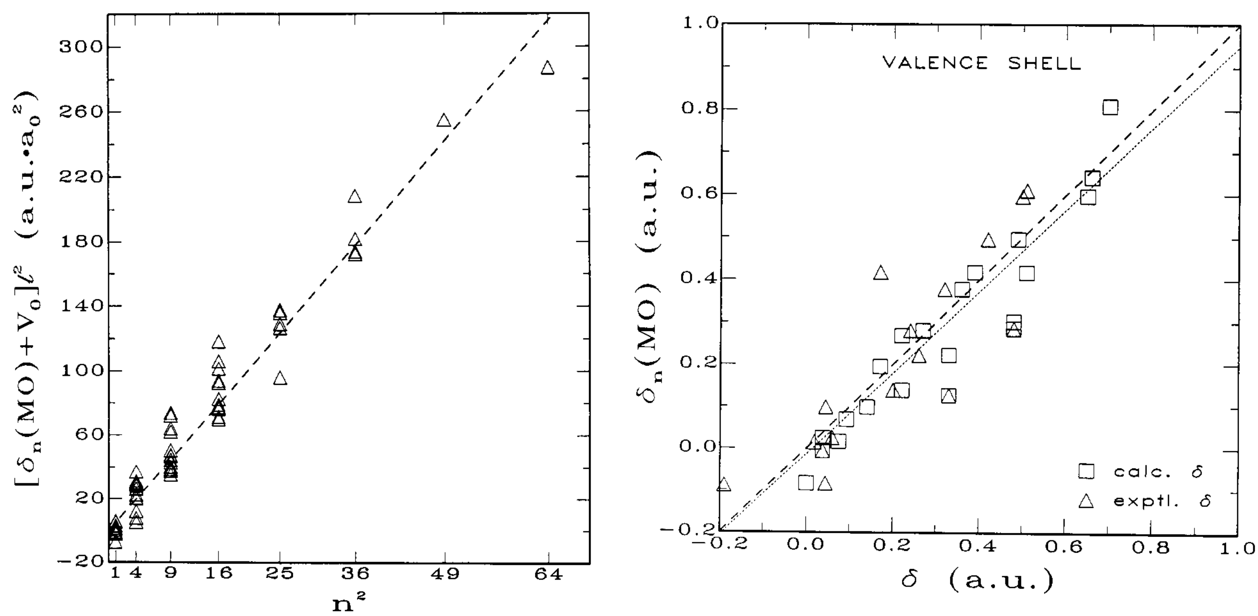

Fig. 5. (a) Correlation of calculated minimal-basis-set occupied and virtual-valence orbital energies (Δ) with effective nodal quantum number n. The predictions of the square-well formula of Eq. (21) are given as a dashed line of slope $\pi^2/2$. (b) Correlation of minimal-basis-set virtual-orbital energies with corresponding experimental and theoretical values for valence-shell ionization, obtained as discussed in the text. Also shown is a least-squares fit to the numerical values (- - -) and a unity correlation line (— — —).

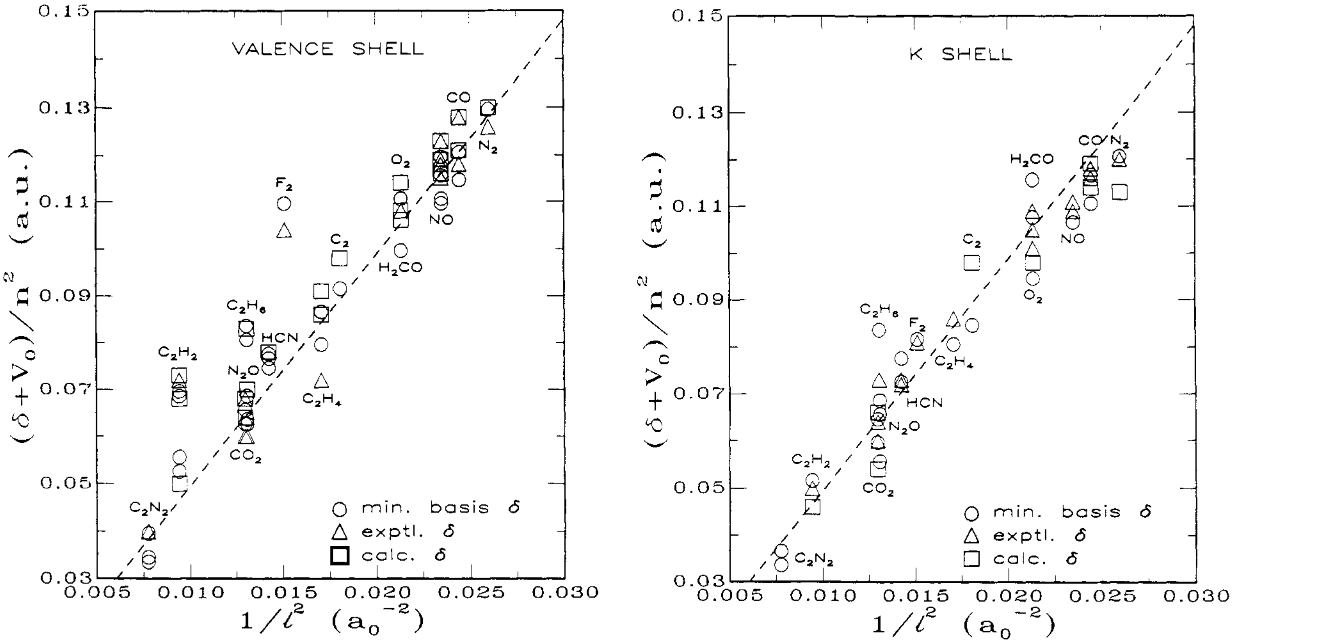

Fig. 6. (a) Correlation of minimal-basis-set (O), experimental (Δ), and calculated (□) valence-shell photoionization resonance energies measured from threshold (δ) with square-well length ($1/l^2$). The dashed line having slope $\pi^2/2$ represents the universal relationship of eq. (21). (b) As in (a) for corresponding K-shell resonances.

ship of eq. (21). With the possible exceptions of the compounds C_2H_2, C_2H_4, and C_2H_6, which are discussed separately below, and F_2, which is subject to configuration-mixing effects [36], the simple formula of eq. (21) based on a cylinderical well model and the adopted well length and depth parameters is seen to give a satisfactory correlation. Figure 6(b) shows corresponding results for K-edge resonances in these compounds employing well length (l) and depth (V_0) parameters which are identical to those employed in Fig. 6(a) [16]. The minimal-basis-set energies reported in Fig. 6(b) (O) refer to unrelaxed molecular-orbital hole-state calculations which have been corrected in an approximate manner to reflect the significant change in the potential experienced by the virtual orbitals accompanying the opening of the K-shell [16].

Figure 6(b) is similar to, but is not identical with, previously reported semi-empirical correlations between K-edge resonance position and bond lengths [5]. The present results employ a correlation with $1/l^2$, whereas the semi-empirical correlation employs the bare internuclear separation R. [5]. When the present results are reported in terms of l rather than $1/l^2$ some curvature is evident in the resulting figure [16], but there is nevertheless a clear basis for a linear relationship between resonance energy and bond length of the type previously reported on semi-empirical grounds [5]. Additionally, there is a general correspondence between the well-depth parameters V_0 employed in the present development and the $Z_A + Z_B$ potential parameter identified in the semi-empirical correlation [5]. The present results not only provide a theoretical basis for the semi-empirically determined correlation [5], but also introduce a number of additional features into the development. Specifically, distinction is made between the effective length l specifying the spatial extent of resonance functions and the molecular bond length, and effective square-well quantum numbers are incorporated in the results of Figs. 5 and 6. Accordingly, the present correlation can deal with cases (e.g., CO_2, N_2O, C_2H_2, ...) in which more than one resonance is present in a given channel, each having a different nodal quantum number.

5. Resonances in simple hydrocarbons

A number of the compounds considered in Figs 5 and 6 include more than one configurational state function or virtual valence orbital above threshold, and, consequently, can exhibit more than one photoionization resonance. The

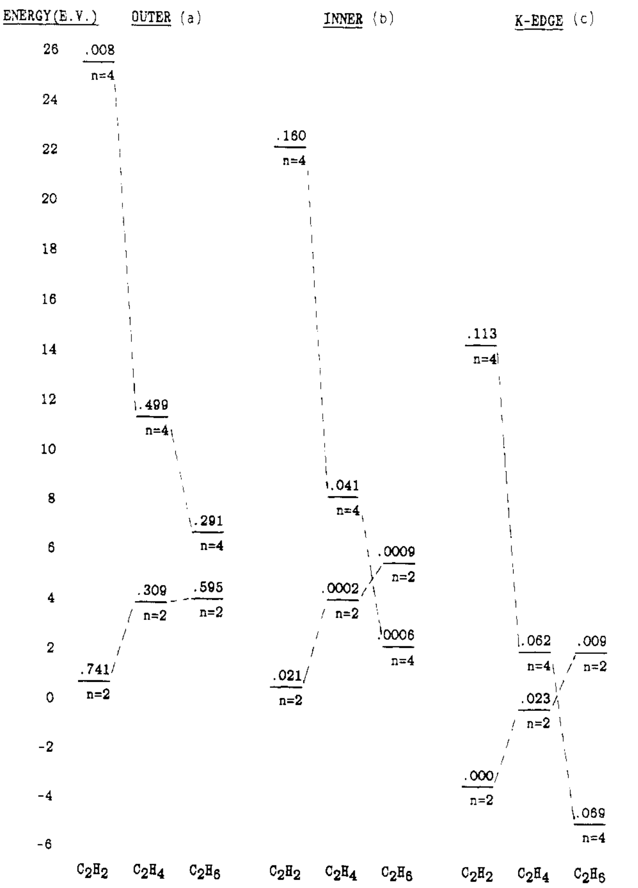

Fig. 7. Mulliken configurational state energies for simple hydrocarbons C_2H_2, C_2H_4, C_2H_6. Values obtained from single-excitation static-exchange calculations employing minimal basis sets for (a) outer-valence, (b) inner-valence, and (c) K-shell excitations. See discussion in the text.

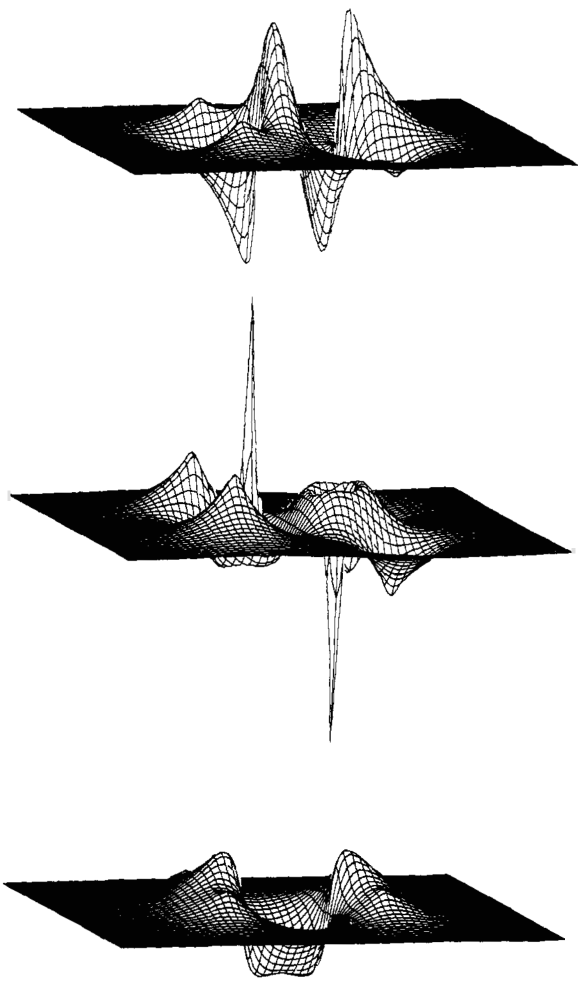

Fig. 8. Occupied $3a_g$ (bottom) and virtual $3b_{1u}$(C–H σ^*), $4b_{1u}$(C–C σ^*) (top) valence orbitals for C_2H_4 calculated in a minimal basis set. Orbital values shown refer to a plane that contains the C–C axis and the four H atoms. Qualitatively similar virtual σ^* orbitals, specifically with respect to C–C and C–H nodal patterns, are obtained for the closely related compounds C_2H_2 and C_2H_6.

simple hydrocarbons C_2H_2, C_2H_4, and C_2H_6 include both C–C and C–H σ^* orbitals in their spectra, and so are of particular interest. Moreover, it is seen from Fig. 5(a) that the valence-shell resonances in these compounds do not satisfy the theoretically derived correlation as closely as do most of the other compounds. In Fig. 7 are shown calculated minimal-basis-set virtual- valence orbital energies in these compounds appropriate for outer-valence, inner-valence, and *K*-edge ionization and excitation employing unrelaxed molecular-orbital hole states in the latter cases. Also shown in the figure are calculated zeroth-order *f* numbers for each of the transitions. The spatial characteristics of these orbitals and of the occupied orbital are depicted in Fig. 8 for the outer-valence $3a_g$ channel in C_2H_4. Qualitatively similar orbital patterns to those of Fig. 8, which depict the classical C–C antibonding nature of the higher lying $4b_{1u}$ orbital and the C–C nonbonding nature of the lower lying $3b_{1u}$ orbital, are also obtained in the cases of the other channels and the two other compounds of interest [15].

Evidently, the C–C σ^* orbitals in Fig. 7 show a smooth decrease in energy with increasing C–C bond length, whereas the C–H σ^* orbitals increase in energy along the hydrocarbon sequence. The former behavior is in accord with the correlation of eq. (21), but is modified by the increasing number of C–H antibonds in the C–C σ^* orbital sequence, which tends to counter, but not overcome, the decrease in energy with increasing C–C bond length. By contrast, the effect of the number of C–H antibonds present dominates the energy behavior of the C–H σ^* orbitals, since they are all nonbonding in the C–C bond, and relatively insensitive to C–C distance. The C–C and C–H σ^* orbital sequences evidently cross or invert order in the inner-valence and *K*-edge spectra. This circumstance has important consequences on the assignment of experimental inner-valence and *K*-edge resonance features [5], in which connection it should be noted that the *K*-shell results of Fig. 8 refer to unrelaxed molecular-orbital or delocalized hole-state calculations. Relaxed-hole-state or local-hole-state minimal-basis-set virtual orbital energy calculations can possibly lead to a different positioning of the *K*-shell energies in Fig. 8.

In Fig. 9 are shown the calculated outer-valence cross sections for C_2H_2 and C_2H_4 obtained following the development of Section 3, in comparison with corresponding measured values [14, 15]. Evidently, there is general agreement between the calculations and the measured values. Similarly good agreement obtains between calculations and measured partial-channel data for the other channels of these compounds [15], although experimental values are apparently not available at present for C_2H_6. Figure 10 summarizes the calculated outer-valence, inner-valence, and *K*-shell $\sigma \rightarrow \sigma$ polarization components in the three compounds of interest, plotted against the kinetic energy of the photoejected for ease of comparison.

The calculated and measured cross sections shown in Figs. 9 and 10 are in accord with the energies and natures of the virtual valence orbitals of Figs. 7 and 8 underlying the resonance features. Referring first to the outer-valence results of Figs. 9 and 10, the absence of a high-lying C–C σ^* resonance in the C_2H_2 cross section is evidently in agreement with the very small f number shown in Fig. 7, as is the presence of the strong near-threshold C–H resonance in this case. Similarly, the appearance of broad combined C–C and C–H σ^* resonances in the outer-valence channels in C_2H_4 and C_2H_6 shown in Fig. 10 is also in satisfactory accord with the energies and f numbers for the underlying orbital transitions reported in Fig. 7. In the inner-valence spectra of Fig. 10, both C–C and C–H resonances are apparent in C_2H_2, again in agreement with the energies and f numbers of the underlying orbital transitions of Fig. 7. The weaker broad features in the cross sections of the other two compounds are not virtual-orbital resonances, however, as discussed in further detail elsewhere [14, 15]. Finally, in the *K*-edge spectra of Fig. 10, as indicated previously, the above threshold resonance in C_2H_6 is found to be of C–H σ^* type rather than of C–C σ^* type, whereas those in the other two compounds are both of C–C σ^* type, in agreement with previous assignments [3–5]. Since the *K*-shell results of Fig. 10(c) refer to unrelaxed molecular-orbital or delocalized-hole calculations only, these assignments should be verified by performing corresponding relaxed-hole-state or local-hole-state calculations of the *K*-edge cross sections.

6. Concluding remarks

Nondegenerate interaction-prepared states having point-group symmetry resolve the infinite degeneracy of molecular electronic continuum states in a conceptually and compu-

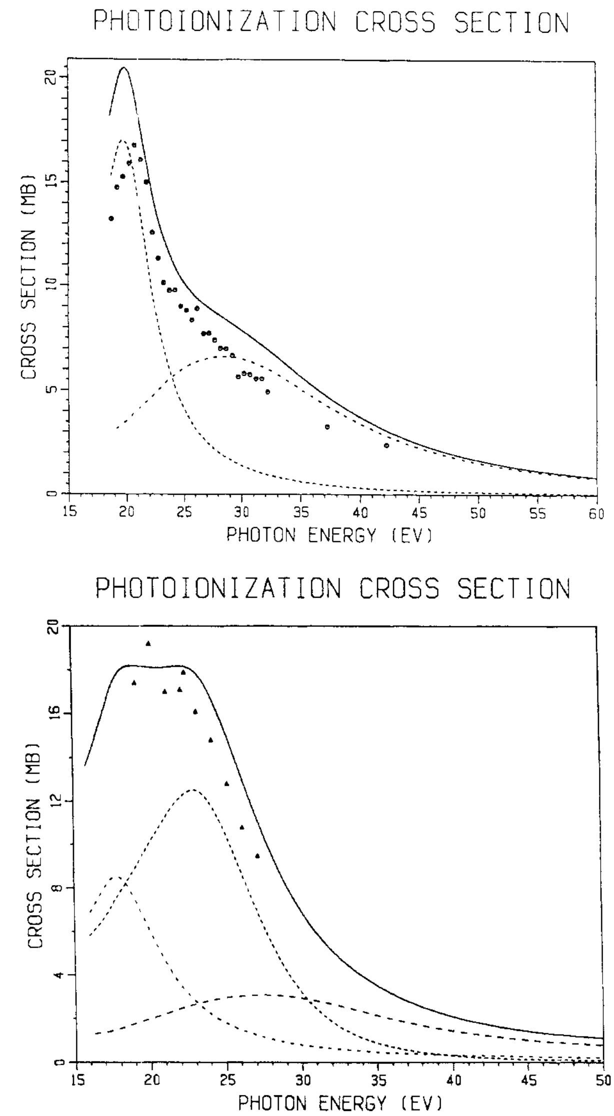

Fig. 9. Single-excitation static-exchange (——) and experimental (○, ▲) partial photoionization cross sections for outer-valence resonance channels in (a) C_2H_2 ($3\sigma_g$) and (b) C_2H_4 ($3a_g$). Dashes refer to polarization components [14, 15].

tationally convenient manner, and provide natural continuum analogues of the more familiar bound electronic states of polyatomic molecules. Partial-wave expansions common to conventional scattering calculations are avoided by calculating the interaction-prepared states directly in L^2 basis sets, rather than by determining the individual degenerate components of which they are comprised. By contrast, conventional scattering wave calculations based on K-matrix, S-matrix, or eigenchannel states generally determine the individual degenerate scattering waves at each energy in order to obtain the desired photoionization cross section. Of course, identical cross sections can be obtained from the L^2 and various scattering-wave developments when identical dynamical approximations are employed.

Feshbach–Fano methods incorporated into the general prepared-state development provide a multi-channel basis-set treatment of complex Hamiltonian spectra. In this way resonance profiles are characterized in terms of zeroth-order states, background cross sections, shifts, widths, and lineshape indices which are generally slowly varying functions of energy, and which can be calculated more conveniently under appropriate conditions than can the more rapidly varying cross section itself. When the various parameters are strong functions of the energy variable, however, a lineshape characterization of the generalized Fano type will be less satisfactory for calculations of molecular photoionization cross sections. Calculations of the partial cross sections of molecules which

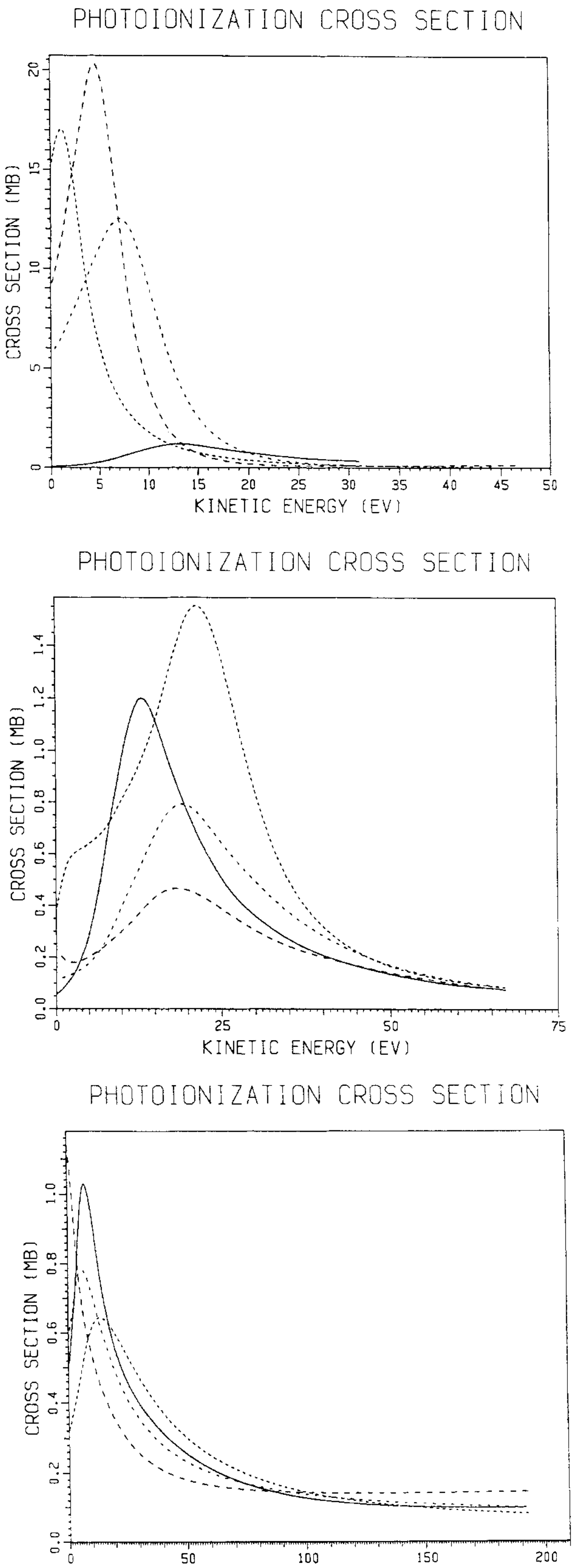

Fig. 10. Resonance-channel photoionization cross sections in C_2H_2 (short dashes), C_2H_4 (longer dashes), and C_2H_6 (longest dashes) obtained in single-excitation, static-exchange approximation for (a) outer-valence, (b) inner-valence, and (c) K-shell ionization. The solid curves refer to C_2 calculations [14, 15].

include shape-resonances employing Mulliken charge-transfer configurations as zeroth-order states result in generally weak and unstructured background cross sections, verifying the long-suspected importance of such states in the formation of molecular shape resonances in specific valence and *K*-shell channels.

A universal correlation between shape resonance energies and molecular geometry is derived on basis of the natures and behaviors of underlying Mulliken states, and on their similarities with the eigenfunctions of a square well. In this way the origins of semi-empirically determined correlations are clarified and elaborated upon. Calculations on simple hydrocarbons illustrate the complexities that can arise in cases of nonlinear compounds which include two resonances in a given channel. The Mulliken-orbital (σ, σ^*) charge-transfer excitations underlying the calculated valence-shell shape resonances are seen to account for the variations in spectral positions and strengths of prominent features in the measured cross sections even in the absence of a simple square-well correlation behavior. Assignments of resonance features in the *K*-edge cross setions of these compounds are complicated by the possible effects of core-hole relaxation and localization, which must be incorporated into the development to achieve definitive theoretical assignments.

Although the cylindrical square-well model employed is seen to be a useful device for correlating the energies of occupied and virtual orbitals in linear molecules with effctive bond lengths, effective potentials, and nodal quantum numbers, some caution is recommended in adoption of the model for other purposes. Specifically, the photoionization cross section of a cylindrical well differs both qualitatively and quantitatively from that of a linear molecule, particularly at the threshold where the former vanishes and the latter can take on a finite value due to the Wigner threshold law associated with a Coulomb potential. Square wells, moreover, exhibit virtual states which strongly affect their photoionization cross sections in the threshold region, whereas molecules apparently do not exhibit such states and threshold effects. Similar cautionary remarks also apply to the uncritical use of spherical- or radial-well models, with and without barriers , in studies of molecular resonance cross sections, particularly when a single angular momentum value is employed to characterize an occupied or scattering state. Such radial-well models would seem to provide somewhat less realistic effective potentials and other characteristics of molecular photoionization resonances than do cylindrical wells, which latter incorporate the anisotropy of the molecular potential in a simple manner.

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